

Secure Charging Session Management in V2G Systems using Blockchain

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The Vehicle-to-Grid Transfer Protocol (V2GTP) under ISO 15118 lacks native encryption, leaving EV charging sessions vulnerable to spoofing, session hijacking, and impersonation attacks. An attacker intercepting communications between the Electric Vehicle (EV) and the Supply Equipment Communication Controller (SECC) can resume or hijack paused charging sessions by impersonating valid parties. We propose a blockchain-based solution utilizing Hyperledger, a permissioned blockchain, to secure V2G charging sessions. Our approach uses cryptographic token rotation for session resumption and records of all session events to an immutable ledger which enables tamper-proof audit trails. The SECC validates resume requests by verifying SHA-256 hashed tokens, with each successful resumption invalidating the previous token and issuing a new one, preventing replay and impersonation attacks. We implemented and evaluated our system, using OMNeT++ for network simulation, and the Hyper ledger Fabric test network. The results show that integrating blockchain effectively prevents session hijacking while maintaining performance suitable for the real world.

Additional Key Words and Phrases: Electric Vehicles, V2G, Blockchain, Security, Charging Sessions, Vehicle to Grid Communication, Resumption Attacks

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1 Introduction

ISO 15118-based Vehicle-to-Grid (V2G) charging relies on message exchanges between EV and SECC [International Organization for Standardization 2014]. However, V2GTP does not provide native encryption at the protocol layer, which makes charging-session control traffic a primary target when an adversary can observe or inject messages on the local charging network. In particular, session pause/resume behaviour can be abused: an attacker who captures session context (or a resumption credential) can attempt to resume or hijack an existing session by impersonating a legitimate EV or SECC, enabling session hijacking, replay, and impersonation attacks.

It is important to distinguish the focus of this work from previous efforts on charging-session reliability mechanisms such as seamless retry. Seamless retry [Watt et al. 2024] is primarily motivated

by improving user experience and operational robustness by automatically re-initiating a failed charging session without requiring user intervention, effectively emulating a manual unplug-replug sequence. In contrast, our work addresses a fundamentally different problem: the prevention of resumption attacks during V2G charging sessions. Prior work shows that weaknesses in V2G communication can be exploited to manipulate charging behaviour and, under some conditions, threaten local grid stability [Zhdanova et al. 2022].

In this work, we investigate the method to harden EV-SECC charging-session management using blockchain, especially, session resumption against hijacking and replay attacks while providing tamper-evident accountability for session events. The scope of this work is limited to securing the EV-SECC communication path for charging-session lifecycle events (start, pause, resume, stop) and, specifically, preventing resumption-based session hijacking. We do not address CPO back-end protocols, billing, energy metering correctness, or power-delivery validation; the goal is secure charging session management rather than energy accounting. To the best of our knowledge, this is the first work to secure V2G charging sessions using blockchain that provisions accountability while preventing invalid session resumptions. We propose a permissioned-blockchain-backed session management mechanism in which (i) the SECC issues a cryptographic resume token and stores only its SHA-256 hash, (ii) each successful resumption rotates the token (revoking the previous one) to prevent replay, and (iii) all session state transitions are recorded on Hyperledger Fabric to provide an immutable audit trail. We prototyped the design and evaluated it using an OMNeT++ based simulation coupled with a Hyperledger Fabric test network.

2 System Model

Our system consists of 3 main entities as shown in Fig. 1: *Electric Vehicle (EV)* initiates charging sessions and stores resume tokens for session resuming; *Supply Equipment Communication Controller (SECC)* manages the session life cycle, validates resume tokens using SHA-256 hashing, and records events to the blockchain; *Charge Point Operator (CPO)* monitors charging sessions and verifies audit trails via read access on the blockchain.

The proposed approach operates under the following assumptions. First, the SECC is a trusted entity with write access to the blockchain. Second, the EV has secure local storage for resume tokens. Third, communication between the EV and the SECC takes place over V2GTP. Accordingly, the threat model and attack surface considered in this work are confined to the V2G communication path.

2.1 Communication Flow

In our work, the session life cycle consists of four states as shown in Fig. 2. When an EV initiates a session, the SECC generates a cryptographic resume token and logs the event to the blockchain. If the session is paused and resumed later, the EV will present that valid resume token. The SECC verifies the token hash, revokes it,

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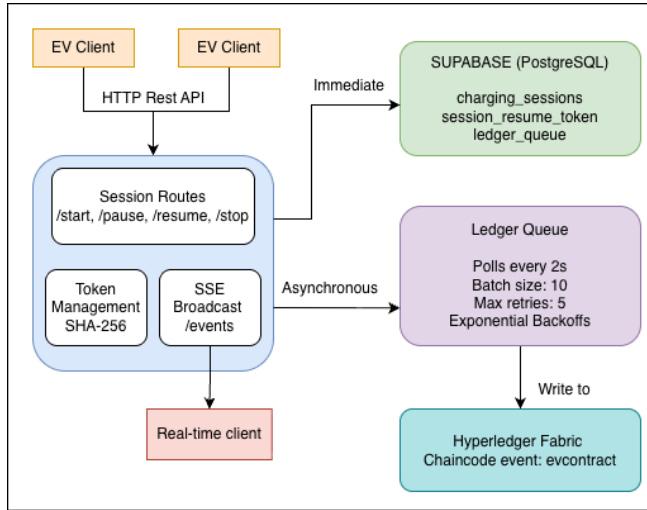


Fig. 1. System Architecture showing EV, SECC, and Blockchain components

and issues a new token to prevent replay attacks. All state transitions are recorded on the Hyperledger Fabric, to provide an immutable audit trail of every event.

3 Implementation

This prototype separates (i) mutable session control/state and (ii) immutable audit logging. Mutable state (session status, token revocation, session expiry) is maintained off-chain to support low-latency decisions during charging, while all session lifecycle events are recorded on a permissioned ledger for accountability and tamper-evident auditing. The on-chain component is implemented on Hyperledger Fabric [Hyperledger Foundation 2024], and we drive the EV-SECC protocol using a minimal OMNeT++ simulation harness. [Varga 2001]

3.1 Permissioned blockchain testbed (Hyperledger Fabric)

We use Hyperledger Fabric because the charging infrastructure is operated by known stakeholders (e.g., CPOs) who can be provisioned with identities and access policies. A permissioned network also avoids the latency, fees, and governance ambiguity of public chains, while preserving an append-only, verifiable history. We deploy our chaincode to the Fabric test-network (two organizations with CAs), mapping *Org1* to the SECC writer role and *Org2* to a CPO auditor role.

3.2 Chaincode: append-only session event ledger

The chaincode (ev-contract) is intentionally *log-oriented*: it does not adjudicate token validity (which must be decided in-line by the SECC), but instead provides an immutable record of what the SECC accepted and when. Each ledger write appends a *SessionEventRecord* containing:

```
<sessionId, evId, eventType, source, payload, txId, timestamp>.
```

Events are stored under keys of the form:

```
sess~{sessionId}~{eventType}~{txId}
```

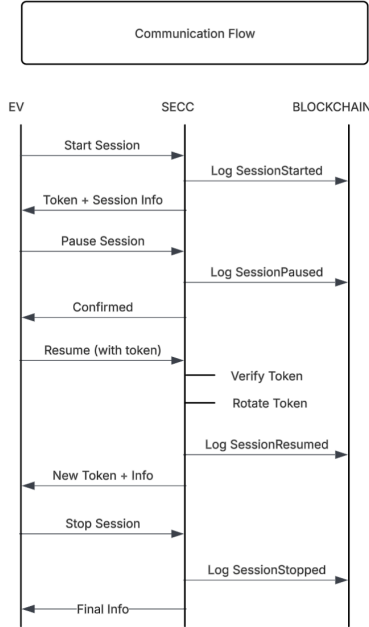


Fig. 2. Communication protocol between EV, SECC, and Blockchain

which enables efficient history retrieval using a prefix range scan over the state database. We deliberately support a LevelDB-compatible history path (range scan + timestamp sort) so the prototype does not depend on CouchDB; a richer JSON query interface is included but treated as optional when CouchDB is enabled. For real-time visibility, each append emits a Fabric chaincode event (session_event), enabling auditor clients to subscribe to session activity without polling.

3.3 SECC service: session control, token hashing, and rotation

The SECC logic is implemented as a TypeScript/Node service that exposes a narrow API matching the session lifecycle (/start, /pause, /resume, /stop). This mirrors how an actual SECC would implement ISO 15118 session management at the application layer, while remaining lightweight enough to integrate with simulation-driven experiments.

3.3.1 Token generation and verification. On session start, the service generates a cryptographically strong resume token using `randomBytes(32)` (256 bits) and encodes it using URL-safe Base64. We choose 256-bit tokens to make online guessing infeasible even under optimistic attacker models, while remaining small enough to pass in protocol messages. The SECC never stores the token in plaintext; it stores only SHA-256(token). This design choice limits blast radius if the SECC database is exposed, since the attacker gains only hashes rather than reusable bearer tokens.

3.3.2 Token rotation on resume. A resume request must present the current resume token. The SECC hashes it, looks up an unrevoked matching hash, and rejects the request if (i) no match exists, (ii)

the token is expired, or (iii) the session is in a terminal state. On a successful resume, the SECC revokes the previous token (sets `revoked_at`) and issues a fresh token with the same session expiry. This makes any intercepted token single-use and directly mitigates replay-based resumption hijacking.

3.3.3 Session expiry and status model. Sessions are created with an expiry time (default 6 hours) and explicit status transitions (active $\rightarrow$ disconnected $\rightarrow$ active, and terminal states stopped/expired). The expiry window is treated as an assumption/parameter: long enough to cover typical “disconnect and return later” use, but bounded to reduce the utility of leaked tokens and to force state cleanup.

3.4 Off-Chain state management and reliability using Supabase with Queued Ledger writes

The SECC service uses an off-chain datastore for *mutable* session state and token revocation (e.g., `charging_sessions`, `session_resume_tokens`). This is intentional: token validation and revocation are on the critical path of resumption and should not depend on blockchain commit latency. To avoid losing audit events when Fabric is temporarily unavailable, the service logs each session event to a `ledger_queue` table and commits to Fabric asynchronously. A background processor polls the queue every 2 seconds, processes up to 10 events per batch, and retries failed submissions up to 5 times with exponential backoff. This choice prioritizes *availability* (sessions can proceed even if Fabric is down) while still converging the audit ledger to the ground truth. Each queued item is marked confirmed after a successful chaincode submission and stores both the Fabric transaction ID and the returned event key. For operator transparency during testing, the service broadcasts lifecycle events and confirmation/failure updates over Server-Sent Events (SSE), which the web UI consumes for live status.

3.5 OMNeT++ protocol driver and integration assumption

For evaluation, we assume a minimal OMNeT++ scenario that repeatedly spawns EV nodes, initiates charging, disconnects for a short interval, resumes, and then terminates the session. The simulated EV maintains the resume token locally and presents it during resumption. The simulated SECC delegates lifecycle handling to the implemented SECC service, which in turn appends immutable events to Fabric. This setup is sufficient to test the security-critical path (resume validation, token rotation and audit logging) without implementing the full ISO 15118 stack.

4 Security Analysis

Our system has in place several security measures against common V2GTP attacks:

Token-based authentication. Resume tokens are generated using `crypto.RandomBytes(32)` to generate 256 bits of cryptographic secure randomness. Tokens are never stored in plain text; the SECC stores only the SHA-256 hash, preventing token theft if the database were ever compromised.

Replay attack prevention. Each successful session resumption will invalidate the current token and issue a new one. This token rotation

Table 1. Security Mechanisms and Mitigated Attacks

Attack	Mitigation
Session Hijacking	Token required to resume
Replay Attack	Token rotation on each resumption
Token Theft	SHA-256 hashing, no plaintext storage
Impersonation	Cryptographic token verification
Audit Tampering	Immutable blockchain ledger

ensures that intercepted tokens cannot be reused by an attacker, as shown in Table 1.

Effectiveness against resumption attacks. All failed session resumption attempts resulting from the presentation of invalid or expired tokens are explicitly logged. In addition, a blockchain-based event history is maintained to provide a complete and tamper-evident audit trail for each charging session. By ensuring verifiable and immutable records of session events, Charging Point Operators gain increased operational transparency and accountability, while reducing exposure to fraud, customer disputes, and liability arising from ambiguous or contested charging transactions. Any session resumption request that exhibits a token hash mismatch is automatically detected and rejected by the SECC, thereby preventing unauthorized or inconsistent session continuation.

5 Conclusion

This work targets a practical weakness in V2G charging: protecting EV-SECC session resumption against hijacking and replay. We introduced a lightweight resume-token scheme that stores only token hashes at the SECC and rotates tokens on every successful resumption, and we paired it with a permissioned-ledger audit trail using Hyperledger Fabric. Future work will focus on integrating a fuller ISO 15118 V2GTP message flow and certificate lifecycle, expanding adversary models (e.g., active man-in-the-middle and race conditions), and validating performance and robustness under larger-scale scenarios and real network conditions. We also plan to evaluate endpoint-hardening options (e.g. secure enclaves/HSM-backed token storage) and stronger bindings between tokens and EV identity to further reduce “risks of first-use wins” from stolen bearer credentials.

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